

Suraj Chougule

+91 8446364651 | surajchougule378@gmail.com | surajc-spec | suraj-chougule | Portfolio

Pune, Maharashtra, India

SUMMARY

Software Development Engineer with **3 production-grade projects** serving **162+ organic users** built **distributed, fault-tolerant systems** achieving **17x throughput gain** ($30 \rightarrow 512$ req/sec) and **1.8M req/hr capacity** using multi-level Redis caching. Experienced in designing **scalable backends, relational databases** (PostgreSQL), and **AI-integrated pipelines**. **1st Place Hackathon Winner**. Strong foundation in **DSA, OOP, Complexity Analysis, Operating Systems**, and **agile development**. Proficient in **Java, Python, C/C++**, and **AWS**. Passionate about **distributed storage, data mining, automation**, and **security**.

EDUCATION

2023 – 2027 B.E. Information Technology, **I²IT Pune** **CGPA: 8.84/10**
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Object-Oriented Design, Computer Networks, Software Engineering, Distributed Systems

TECHNICAL EXPERIENCE

NoteShare — Production-Scale Academic Platform [\[Live\]](#) [\[GitHub\]](#)

- Built a **distributed, fault-tolerant** MERN backend serving **162+ organic users** with **17x throughput gain** ($30 \rightarrow 512$ req/sec) and **1.8M req/hr capacity** demonstrating **Scalability, Distributed Systems**, and **High Availability**
- Implemented **multi-level caching** (**L1 + L2 Redis**) with **fault-tolerant fallback** to database optimized query latency to **$O(\log n)$** using **compound indexes** and **.lean()** queries; reduced database load by **90%**
- Designed secure **RBAC** system with **PDF watermarking, blocked shortcuts**, and **Android app** (**Capacitor + FLAG_SECURE**) demonstrating **end-to-end ownership** of a **multi-tiered system**
- Implemented **production observability** via **Google Analytics** (**27-min avg session time**), **structured error handling**, and **diagnostic logging** **Agile** development with continuous monitoring
- Stack:** React, Node.js, Express, MongoDB, Redis, Cloudflare R2, JWT

Aura AI — Multilingual AI Image Generator [\[Live\]](#) [\[GitHub\]](#)

- Built **distributed AI pipeline** supporting **11 Indian languages** with **100% language accessibility** — demonstrating **Customer Obsession** through inclusive design
- Integrated **Gemini API** (translation) and **Cloudflare AI** (generation) across **multi-tiered architecture** with ImageKit storage and MongoDB — managing concurrent requests with **fault-tolerant** fallback paths
- Implemented **JWT authentication, protected routes**, user dashboard, and **rate limiting** — ensuring secure, **fault-tolerant** concurrent multi-user access
- Stack:** React, Node.js, Express, MongoDB, Gemini API, Cloudflare AI, ImageKit, JWT

Expense Splitter — Smart Group Settlement System [\[GitHub\]](#)

- Designed debt settlement system using **greedy graph optimization** reducing **50 transactions to 3-5 settlements** ($O(N^2) \rightarrow O(N-1)$) — demonstrating strong **Algorithms, Data Structures**, and **Complexity Analysis** skills
- Applied **Strategy Pattern** for Equal/Exact/Percentage/Adjustment splits — showcasing **Object-Oriented Design** and extensible architecture
- Integrated **Gemini Vision API** for **OCR receipt scanning** auto-extracting items, amounts, tax, and GST reducing manual entry by **80%** and **Delivering Results** for end users
- Built interactive dashboard with **Recharts** and **PostgreSQL relational database** schema demonstrating **Relational Databases, Multi-tiered systems**, and data persistence
- Stack:** React, Node.js, PostgreSQL, Gemini Vision API, Redis, JWT, Recharts

LEADERSHIP EXPERIENCE

Vice Chairperson — IT Students' Association (ITSA), I²IT Pune July 2025 – Aug 2026

- Led and mentored a team of 50 members, fostering a culture of **innovation, collaboration**, and **professional commitment** while demonstrating **Ownership, influence**, and **Bias for Action**
- Articulated roadmaps, **created** onboarding processes, and **reported** progress to faculty; **adept** at **planning, interviewing**, and **supporting** teams to deliver 200+ participant events
- Competently managed ambiguous requirements, **wrote** documentation, and made data-driven **predictions** to **influence** decisions and **deliver results** at scale

ACHIEVEMENTS

- 1st Place** — Dhruva Nexus Hackathon
- NoteShare** — 162+ organic users, 27-min avg session time, zero marketing spend

TECHNICAL SKILLS

Languages	Java, Python, C/C++, JavaScript, SQL, HTML5, CSS3, Perl
Frontend	React.js, Vite, Tailwind CSS, React Router, Recharts
Backend	Node.js, Express.js, REST APIs, Microservices
Databases	MongoDB, PostgreSQL (Relational Databases), Redis (Caching, Indexing)
Cloud	AWS (Amazon Web Services), Cloudflare R2, ImageKit, Vercel, Render
AI/ML	Gemini API, Cloudflare AI
Concepts	Distributed Systems, Distributed Storage, Scalability, Fault-Tolerance, High Availability, OOP, Object-Oriented Design, Design Patterns (Strategy), DSA, Complexity Analysis, Multi-level Caching, JWT Auth, RBAC, Agile, CI/CD, Performance Monitoring, Relational Databases, Multi-Tiered Systems, Optimization, Linear Programming, Mathematics, Data Mining, Automation, Security, Testing (Postman)